Oracle AI Database 26ai Free

Backup, Restore & Migration Notes

End-to-end guide for moving a Container Database (CDB) **FREE** with three Pluggable Databases (**FREEPDB1**, **HRDB**, **POSDB**) from one server/instance to another, plus routine backups, automation scripts, verification and troubleshooting.

Item	Value used in these notes
Source server	**srcserver** (the machine you are moving FROM)
Target server	**tgtserver** (the machine you are moving TO)
CDB / SID	FREE (service name: FREE)
PDBs	FREEPDB1, HRDB, POSDB (service names: freepdb1, hrdb, posdb)
Listener port	1521
Linux backup folder	/u01/backup (change to any folder with enough space)
Windows backup folder	D:\backup
Passwords	Shown as <pwd> - replace with your real password

IMPORTANT: Always test the restore on the target before you delete or change anything on the source. A backup you have never restored is only a hope, not a backup.

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1. Understand Your Setup

Oracle 26ai uses the **multitenant** architecture. The CDB (called **FREE**) is the container that owns the instance, memory, control files, redo logs and the root (CDB$ROOT). Each PDB is a self-contained database with its own users, tables and datafiles, plugged into the CDB.

```
CDB: FREE  (instance, SGA, control files, redo logs, undo, CDB$ROOT)
 |-- PDB$SEED   (read-only template used to create new PDBs)
 |-- FREEPDB1   (default PDB created by the installer)
 |-- HRDB       (your PDB)
 |-- POSDB      (your PDB)
```

1.1 Free edition limits to remember

- Uses at most 2 CPU threads and 2 GB of RAM for the database.
- User data is limited to about **12 GB** in total - the target must stay within this, so check sizes before importing.
- The CDB name is fixed as **FREE** - you cannot create a second CDB with another name on the same server.
- Install the **same release** (same 26ai version/release update) on the target, especially for RMAN, unplug/plug and cloning.

1.2 Inspect the source - run these first

Connect to the root as SYSDBA:

```
# Linux
export ORACLE_SID=FREE
sqlplus / as sysdba

REM Windows (Command Prompt)
set ORACLE_SID=FREE
sqlplus / as sysdba
```

```
-- Version and edition
SELECT banner_full FROM v$version;

-- CDB info, log mode, DBID (write the DBID down - needed for RMAN restore)
SELECT name, dbid, log_mode, cdb, open_mode FROM v$database;

-- All PDBs and their state
SHOW PDBS
SELECT con_id, name, open_mode, guid FROM v$pdbs;

-- Character set (source and target must be compatible)
SELECT parameter, value FROM nls_database_parameters
 WHERE parameter IN ('NLS_CHARACTERSET','NLS_NCHAR_CHARACTERSET');

-- Where every file lives
SELECT con_id, file_name, ROUND(bytes/1024/1024) mb FROM cdb_data_files ORDER BY con_id;
SELECT con_id, file_name FROM cdb_temp_files;
SELECT member FROM v$logfile;
SELECT name FROM v$controlfile;
SHOW PARAMETER spfile
SHOW PARAMETER db_create_file_dest
SHOW PARAMETER db_recovery_file_dest

-- Size per PDB (GB)
SELECT c.name, ROUND(SUM(d.bytes)/1024/1024/1024,2) gb
  FROM cdb_data_files d JOIN v$containers c ON c.con_id=d.con_id
 GROUP BY c.name ORDER BY 1;

-- Is local undo on? (needed for hot cloning)
SELECT property_value FROM database_properties WHERE property_name='LOCAL_UNDO_ENABLED';

-- Any encryption (TDE)? If rows return, you must also move the wallet/keys
SELECT * FROM v$encryption_wallet;
SELECT con_id, ts#, encryptionalg FROM v$encrypted_tablespaces;
```

Inside each PDB, list your application (non-Oracle) schemas:

```
ALTER SESSION SET CONTAINER = HRDB;
SELECT username, default_tablespace, account_status
  FROM dba_users WHERE oracle_maintained = 'N' ORDER BY 1;
SELECT owner, COUNT(*) FROM dba_objects
 WHERE owner IN (SELECT username FROM dba_users WHERE oracle_maintained='N')
 GROUP BY owner;
-- Repeat for POSDB and FREEPDB1
```

> **TIP:** Typical Linux file locations for Free: datafiles under /opt/oracle/oradata/FREE/ and one sub-folder per PDB (e.g. /opt/oracle/oradata/FREE/HRDB/). On Windows they are under C:\app\<user>\product\26ai\oradata\FREE\. Always trust the query output above rather than these defaults.

2. Which Method Should I Use?

Method	What it moves	Best for	Source downtime	Main requirements
A. Data Pump (expdp/impdp)	Schemas / full PDB contents (logical)	Different version, OS, or char set; moving only some schemas; cleaning up	None (consistent export with flashback_time)	Directory object; target PDB created first
B. Unplug / plug (.pdb)	Whole PDB with all its datafiles	Moving complete PDBs to another CDB of same release	PDB closed during unplug (or clone it first)	Same release; compatible char set and options
C. Remote hot clone	Whole PDB, copied over the network	Easiest PDB move when both servers can talk	None (source stays open)	Archivelog mode, local undo, DB link, same release
D. RMAN backup/restore	Entire CDB with all 3 PDBs	Full server move or disaster recovery; real backups	None for backup (archivelog mode)	Same release and OS family; same DBID restored
E. Cold copy	Entire CDB files	Simple lab / same paths, same OS	Full shutdown	Identical release, OS and paths

Recommendation for your setup

- **Moving everything to a new machine** (same 26ai Free release): use **Method D (RMAN)** - one backup carries FREE + FREEPDB1 + HRDB + POSDB.
- **Moving just HRDB and POSDB** into an existing 26ai on another machine: use **Method C** (if network connects) or **Method B**.
- **Different version, OS, or you only need application data**: use **Method A (Data Pump)**.
- **Best practice:** take a Data Pump export of each PDB *in addition* to RMAN - it is your safety net if a physical restore fails.

3. Pre-Migration Checks

3.1 Source checklist

- Record version (`v$version`), DBID, character sets, PDB list, file paths and sizes (section 1.2).
- Make sure you have free disk space for the backup: roughly the used size of the data (compressed RMAN is often 20-40% of that).
- Note any DIRECTORY objects, DB links, scheduler jobs, and the APEX / ORDS version if used.
- Note the SYS / SYSTEM / PDBADMIN passwords.
- Stop application connections during the final export or cut-over, to avoid lost transactions.

```
-- Objects that move with the data but point to OS paths / other DBs
SELECT directory_name, directory_path FROM dba_directories;   -- in each PDB
SELECT owner, db_link, host FROM dba_db_links;
SELECT owner, job_name, enabled FROM dba_scheduler_jobs WHERE owner NOT IN ('SYS','SYSTEM');
-- APEX version (if installed in the PDB)
SELECT version_no FROM apex_release;
```

3.2 Target checklist

- Install Oracle AI Database 26ai Free - **same release** as the source.
- Confirm the listener is running: `lsnrctl status`.
- Create the backup folder and give the oracle OS user read/write permission.
- Make sure the target character set matches the source (or is a superset for Data Pump).
- Check free space for datafiles plus the 12 GB user data limit.

```
# Linux - service control script (name follows the release; check with: ls /etc/init.d/oracle-free*)
sudo /etc/init.d/oracle-free-26ai status
sudo /etc/init.d/oracle-free-26ai start

# Create backup folders (Linux)
sudo mkdir -p /u01/backup/dp /u01/backup/rman /u01/backup/pdb
sudo chown -R oracle:oinstall /u01/backup

REM Windows - check services in services.msc: OracleServiceFREE and the TNS listener service
mkdir D:\backup\dp D:\backup\rman D:\backup\pdb
```

3.3 Copying files between servers

```
# Linux to Linux
scp -r /u01/backup/rman oracle@tgtserver:/u01/backup/
rsync -avP /u01/backup/dp/ oracle@tgtserver:/u01/backup/dp/

# Windows: copy over a shared folder, USB disk, or
robocopy D:\backup \\tgtserver\D$\backup /E /Z /R:2 /W:5
```

IMPORTANT: Copy dump and backup files in **binary** mode. If using FTP/WinSCP make sure it is not set to text/ASCII mode, otherwise the files become corrupt.

4. Method A - Data Pump Export / Import per PDB

Data Pump is a **logical** backup: it writes table data and object definitions into .dmp files. It is the most flexible method and works across versions, operating systems and paths. You run it **once per PDB** (FREEPDB1, HRDB, POSDB). Never export from CDB$ROOT for application data.

A.1 On the SOURCE - create a directory object in each PDB

```
sqlplus / as sysdba

ALTER SESSION SET CONTAINER = HRDB;
CREATE OR REPLACE DIRECTORY dp_dir AS '/u01/backup/dp';     -- Windows: 'D:\backup\dp'
GRANT READ, WRITE ON DIRECTORY dp_dir TO system;

ALTER SESSION SET CONTAINER = POSDB;
CREATE OR REPLACE DIRECTORY dp_dir AS '/u01/backup/dp';
GRANT READ, WRITE ON DIRECTORY dp_dir TO system;

ALTER SESSION SET CONTAINER = FREEPDB1;
CREATE OR REPLACE DIRECTORY dp_dir AS '/u01/backup/dp';
GRANT READ, WRITE ON DIRECTORY dp_dir TO system;
```

IMPORTANT: The folder must already exist on the OS and be writable by the Oracle software owner. CREATE DIRECTORY does not create the folder.

A.2 Export - option 1: FULL export of each PDB

Run from the OS prompt (not inside SQL*Plus). Connect to the PDB **service name**:

```
expdp system/<pwd>@//localhost:1521/hrdb full=y directory=dp_dir \
      dumpfile=hrdb_full_%U.dmp logfile=hrdb_full_exp.log filesize=2G \
      flashback_time=systimestamp

expdp system/<pwd>@//localhost:1521/posdb full=y directory=dp_dir \
      dumpfile=posdb_full_%U.dmp logfile=posdb_full_exp.log filesize=2G \
      flashback_time=systimestamp

expdp system/<pwd>@//localhost:1521/freepdb1 full=y directory=dp_dir \
      dumpfile=freepdb1_full_%U.dmp logfile=freepdb1_full_exp.log filesize=2G \
      flashback_time=systimestamp
```

On Windows put each command on one line (remove the \ line breaks, or use ^).

A.3 Export - option 2: SCHEMA export (cleaner, recommended)

Exports only your application users (e.g. HR, HRAPP, POS). Fewer errors on import because Oracle-owned objects are skipped.

```
expdp system/<pwd>@//localhost:1521/hrdb schemas=HR,HRAPP directory=dp_dir \
      dumpfile=hrdb_schemas_%U.dmp logfile=hrdb_schemas_exp.log flashback_time=systimestamp

expdp system/<pwd>@//localhost:1521/posdb schemas=POS directory=dp_dir \
      dumpfile=posdb_schemas_%U.dmp logfile=posdb_schemas_exp.log flashback_time=systimestamp
```

Useful Data Pump parameters

Parameter	Purpose
full=y / schemas=A,B / tables=A.T1	What to export
flashback_time=systimestamp	Consistent snapshot of all tables at one point in time
filesize=2G with %U in dumpfile	Splits into multiple files (easier to copy)
exclude=statistics	Skip stats; regather on target (faster import)

Parameter	Purpose
content=metadata_only	Structure only, no rows
estimate_only=y	Shows the expected size without exporting
parfile=exp_hrdb.par	Put all parameters in a file (avoids quoting issues on Windows)

Always open the .log file and confirm the last line says **successfully completed**.

A.4 Copy the dump files to the target

```
scp /u01/backup/dp/*.dmp /u01/backup/dp/*.log oracle@tgtserver:/u01/backup/dp/
```

A.5 On the TARGET - create the PDBs (HRDB, POSDB)

FREEPDB1 already exists after a fresh install, so only HRDB and POSDB need creating. They are created from PDB$SEED:

```
sqlplus / as sysdba

-- Linux paths (adjust to your target)
CREATE PLUGGABLE DATABASE hrdb ADMIN USER pdbadmin IDENTIFIED BY "<pwd>"
  FILE_NAME_CONVERT = ('/opt/oracle/oradata/FREE/pdbseed/',
                       '/opt/oracle/oradata/FREE/HRDB/');

CREATE PLUGGABLE DATABASE posdb ADMIN USER pdbadmin IDENTIFIED BY "<pwd>"
  FILE_NAME_CONVERT = ('/opt/oracle/oradata/FREE/pdbseed/',
                       '/opt/oracle/oradata/FREE/POSDB/');

ALTER PLUGGABLE DATABASE hrdb  OPEN;
ALTER PLUGGABLE DATABASE posdb OPEN;
ALTER PLUGGABLE DATABASE ALL SAVE STATE;    -- PDBs auto-open after restart
SHOW PDBS
```

Find the real seed path with `SELECT file_name FROM cdb_data_files WHERE con_id=2;`

A.6 Create tablespaces + directory in each target PDB

If your schemas use their own tablespaces (e.g. HR_DATA, POS_DATA), create them first, or a full import will try to create them using the *source* file paths:

```
ALTER SESSION SET CONTAINER = HRDB;
CREATE TABLESPACE hr_data DATAFILE '/opt/oracle/oradata/FREE/HRDB/hr_data01.dbf'
  SIZE 200M AUTOEXTEND ON NEXT 50M MAXSIZE 4G;
CREATE OR REPLACE DIRECTORY dp_dir AS '/u01/backup/dp';
GRANT READ, WRITE ON DIRECTORY dp_dir TO system;

-- Repeat for POSDB and FREEPDB1 as needed
```

A.7 Import

```
# Full import
impdp system/<pwd>@//localhost:1521/hrdb full=y directory=dp_dir \
      dumpfile=hrdb_full_%U.dmp logfile=hrdb_full_imp.log

# Schema import (recommended)
impdp system/<pwd>@//localhost:1521/hrdb schemas=HR,HRAPP directory=dp_dir \
      dumpfile=hrdb_schemas_%U.dmp logfile=hrdb_schemas_imp.log

impdp system/<pwd>@//localhost:1521/posdb schemas=POS directory=dp_dir \
      dumpfile=posdb_schemas_%U.dmp logfile=posdb_schemas_imp.log

# Common extras:
#   remap_tablespace=USERS:HR_DATA          put objects in a different tablespace
#   remap_schema=HR:HR_TEST                 import under another user name
#   remap_datafile="'/old/x.dbf':'/new/x.dbf'"  change datafile paths on full import
#   table_exists_action=replace|append|truncate|skip
```

TIP: On a FULL import into a new PDB you will see errors such as ORA-31684 'object already exists' for SYSTEM/Oracle-owned objects. These are normal and can be ignored. Look for errors on **your** schemas only.

Then run the post-migration steps (section 9) and verification (section 10).

5. Method B - Unplug / Plug a PDB (.pdb archive)

Unplugging writes the PDB's metadata **and** all its datafiles into one compressed **.pdb** archive file. You copy that file to the target CDB and plug it in. Both CDBs must run the same release.

> **IMPORTANT:** UNPLUG removes the PDB from service on the source. To keep the source PDB working, first make a local clone (step B.1) and unplug the clone instead.

B.1 (Optional but recommended) Clone first so the source keeps running

```
sqlplus / as sysdba
CREATE PLUGGABLE DATABASE hrdb_mig FROM hrdb
  FILE_NAME_CONVERT = ('/opt/oracle/oradata/FREE/HRDB/',
                       '/opt/oracle/oradata/FREE/HRDB_MIG/');
```

B.2 Unplug into a .pdb archive

```
ALTER PLUGGABLE DATABASE hrdb_mig CLOSE IMMEDIATE;
ALTER PLUGGABLE DATABASE hrdb_mig UNPLUG INTO '/u01/backup/pdb/hrdb.pdb';
DROP PLUGGABLE DATABASE hrdb_mig INCLUDING DATAFILES;   -- removes the temporary clone

-- Repeat for POSDB (posdb_mig -> /u01/backup/pdb/posdb.pdb)
```

If you are unplugging the real PDB instead of a clone: close it, unplug it, and later either `DROP PLUGGABLE DATABASE hrdb KEEP DATAFILES;` or plug it back in on the source.

B.3 Copy and check compatibility on the target

```
scp /u01/backup/pdb/*.pdb oracle@tgtserver:/u01/backup/pdb/

-- On target, in CDB$ROOT
SET SERVEROUTPUT ON
DECLARE
  ok BOOLEAN;
BEGIN
  ok := DBMS_PDB.CHECK_PLUG_COMPATIBILITY(
          pdb_descr_file => '/u01/backup/pdb/hrdb.pdb',
          pdb_name       => 'HRDB');
  DBMS_OUTPUT.PUT_LINE(CASE WHEN ok THEN 'COMPATIBLE' ELSE 'NOT COMPATIBLE' END);
END;
/

SELECT name, cause, type, message, status
  FROM pdb_plug_in_violations WHERE status <> 'RESOLVED';
```

Type **ERROR** rows must be fixed before plugging in. **WARNING** rows are usually acceptable.

B.4 Plug in

```
-- Make sure the destination folder exists first
CREATE PLUGGABLE DATABASE hrdb USING '/u01/backup/pdb/hrdb.pdb'
  CREATE_FILE_DEST = '/opt/oracle/oradata/FREE/HRDB';

-- Alternative if you prefer explicit names:
--   ... USING '/u01/backup/pdb/hrdb.pdb'
--       FILE_NAME_CONVERT = ('/opt/oracle/oradata/FREE/HRDB_MIG/',
--                            '/opt/oracle/oradata/FREE/HRDB/');

ALTER PLUGGABLE DATABASE hrdb OPEN;
ALTER PLUGGABLE DATABASE hrdb SAVE STATE;
SELECT name, open_mode, restricted FROM v$pdbs;
```

IMPORTANT: If the PDB opens in RESTRICTED mode (RESTRICTED = YES), query PDB_PLUG_IN_VIOLATIONS - it usually means a patch/version mismatch. Run `$ORACLE_HOME/OPatch/datapatch -verbose`, then close and reopen the PDB. Add **AS CLONE** to CREATE PLUGGABLE DATABASE if a PDB with the same GUID already exists in the target CDB.

6. Method C - Remote Hot Clone over a Database Link

The target CDB pulls a complete copy of a PDB straight from the source over the network. The source PDB stays open and usable. This is the easiest way to move individual PDBs.

C.1 Source prerequisites

```
sqlplus / as sysdba
SELECT log_mode FROM v$database;     -- must be ARCHIVELOG for a hot clone
SELECT property_value FROM database_properties
 WHERE property_name = 'LOCAL_UNDO_ENABLED';   -- must be TRUE

-- Enable ARCHIVELOG if needed (short restart)
SHUTDOWN IMMEDIATE;
STARTUP MOUNT;
ALTER DATABASE ARCHIVELOG;
ALTER DATABASE OPEN;
ALTER PLUGGABLE DATABASE ALL OPEN;

-- Common user used by the clone link
CREATE USER c##clone_user IDENTIFIED BY "<pwd>" CONTAINER = ALL;
GRANT CREATE SESSION, CREATE PLUGGABLE DATABASE TO c##clone_user CONTAINER = ALL;
```

Open port 1521 on the source firewall (Linux: `sudo firewall-cmd --add-port=1521/tcp --permanent; sudo firewall-cmd --reload`; Windows: add an inbound rule for TCP 1521).

C.2 On the target - create the link and clone

```
sqlplus / as sysdba

CREATE DATABASE LINK clone_link
  CONNECT TO c##clone_user IDENTIFIED BY "<pwd>"
  USING '//srcserver:1521/FREE';

SELECT * FROM dual@clone_link;          -- test the link

CREATE PLUGGABLE DATABASE hrdb FROM hrdb@clone_link
  FILE_NAME_CONVERT = ('/opt/oracle/oradata/FREE/HRDB/',
                       '/opt/oracle/oradata/FREE/HRDB/');

CREATE PLUGGABLE DATABASE posdb FROM posdb@clone_link
  FILE_NAME_CONVERT = ('/opt/oracle/oradata/FREE/POSDB/',
                       '/opt/oracle/oradata/FREE/POSDB/');

ALTER PLUGGABLE DATABASE hrdb  OPEN;
ALTER PLUGGABLE DATABASE posdb OPEN;
ALTER PLUGGABLE DATABASE ALL SAVE STATE;

DROP DATABASE LINK clone_link;          -- clean up afterwards
```

If the target already has a FREEPDB1 and you want the source's FREEPDB1 contents, either clone it under a new name (`CREATE PLUGGABLE DATABASE freepdb1_src FROM freepdb1@clone_link ...`) or drop the target's empty FREEPDB1 first.

> **TIP:** For a near-zero downtime cut-over you can create a **refreshable clone** (`... FROM hrdb@clone_link REFRESH MODE MANUAL;`), refresh it with `ALTER PLUGGABLE DATABASE hrdb REFRESH;` while closed, and on cut-over day run `ALTER PLUGGABLE DATABASE hrdb REFRESH MODE NONE;` then open it.

7. Method D - RMAN Full CDB Backup & Restore

RMAN (Recovery Manager) takes a **physical** backup of the whole CDB - root, seed and all three PDBs - in one go. This is the proper production-style backup and the best way to move the entire database to a new server.

D.1 Enable ARCHIVELOG mode on the source (once)

```
sqlplus / as sysdba
ARCHIVE LOG LIST
-- Recovery area (location + size limit)
ALTER SYSTEM SET db_recovery_file_dest_size = 20G SCOPE=BOTH;
ALTER SYSTEM SET db_recovery_file_dest = '/opt/oracle/fra' SCOPE=BOTH;   -- folder must exist

SHUTDOWN IMMEDIATE;
STARTUP MOUNT;
ALTER DATABASE ARCHIVELOG;
ALTER DATABASE OPEN;
ALTER PLUGGABLE DATABASE ALL OPEN;
ALTER PLUGGABLE DATABASE ALL SAVE STATE;
ARCHIVE LOG LIST          -- should now say: Archive Mode
```

D.2 Configure RMAN and take the backup

```
rman target /

SHOW ALL;
CONFIGURE CONTROLFILE AUTOBACKUP ON;
CONFIGURE CONTROLFILE AUTOBACKUP FORMAT FOR DEVICE TYPE DISK TO '/u01/backup/rman/cf_%F';
CONFIGURE RETENTION POLICY TO RECOVERY WINDOW OF 7 DAYS;

RUN {
  ALLOCATE CHANNEL c1 DEVICE TYPE DISK FORMAT '/u01/backup/rman/db_%d_%T_%U';
  BACKUP AS COMPRESSED BACKUPSET DATABASE PLUS ARCHIVELOG;
  BACKUP CURRENT CONTROLFILE FORMAT '/u01/backup/rman/ctl_%d_%T_%U';
  BACKUP SPFILE              FORMAT '/u01/backup/rman/spf_%d_%T_%U';
  RELEASE CHANNEL c1;
}

LIST BACKUP SUMMARY;
RESTORE DATABASE VALIDATE;      -- proves the backup is readable (does not restore)
```

Backing up a single PDB is also possible: `BACKUP PLUGGABLE DATABASE hrdb, posdb;`

D.3 Collect everything needed on the target

File	Linux location (typical)	Windows location (typical)
RMAN backup pieces	/u01/backup/rman/*	D:\backup\rman*
Password file	$ORACLE_HOME/dbs/orapwFREE (or $(orabaseconfig)/dbs)	%ORACLE_HOME%\database\PWDFREE.ora
DBID	From SELECT dbid FROM v$database	Same
Wallet (only if TDE is used)	WALLET_ROOT folder	WALLET_ROOT folder
Network files (optional)	$ORACLE_HOME/network/admin/*.ora	%ORACLE_HOME%\network\admin*.ora

Find the real config folder with `orabaseconfig` (read-only Oracle homes keep dbs/ under ORACLE_BASE).

D.4 Prepare the target

- Install 26ai Free (same release). The installer creates its own empty FREE database.
- Stop that database and **move its files aside** (do not delete until the restore works).

- Create the same folder structure as the source (datafile folders, audit folder, FRA folder).
- Copy the backup pieces and password file into place.

```
# Linux (as oracle)
sudo /etc/init.d/oracle-free-26ai stop
mv /opt/oracle/oradata/FREE /opt/oracle/oradata/FREE_orig_install
mkdir -p /opt/oracle/oradata/FREE/HRDB /opt/oracle/oradata/FREE/POSDB \
         /opt/oracle/oradata/FREE/FREEPDB1 /opt/oracle/oradata/FREE/pdbseed \
         /opt/oracle/fra /opt/oracle/admin/FREE/adump
cp /u01/backup/orapwFREE $ORACLE_HOME/dbs/orapwFREE
```

D.5 Restore and recover

```
export ORACLE_SID=FREE          # Windows: set ORACLE_SID=FREE
rman target /

SET DBID <your_dbid>;
STARTUP NOMOUNT FORCE;          -- starts with a dummy parameter file, this is expected

RESTORE SPFILE FROM '/u01/backup/rman/spf_FREE_<date>_<id>';
STARTUP FORCE NOMOUNT;          -- now using the restored spfile

RESTORE CONTROLFILE FROM '/u01/backup/rman/ctl_FREE_<date>_<id>';
ALTER DATABASE MOUNT;

CATALOG START WITH '/u01/backup/rman/' NOPROMPT;   -- register pieces at the new location

RESTORE DATABASE;
RECOVER DATABASE;
-- RMAN-06054 'media recovery requesting unknown archived log' at the end is NORMAL:
-- it means all available logs were applied.

ALTER DATABASE OPEN RESETLOGS;
ALTER PLUGGABLE DATABASE ALL OPEN;
ALTER PLUGGABLE DATABASE ALL SAVE STATE;
```

D.6 If the target folder paths are different

```
RUN {
  SET NEWNAME FOR DATABASE TO '/data/oradata/FREE/%b';   -- %b = original file name
  RESTORE DATABASE;
  SWITCH DATAFILE ALL;
  SWITCH TEMPFILE ALL;
  RECOVER DATABASE;
}
-- Then rename redo logs before opening (in SQL*Plus, database mounted):
ALTER DATABASE RENAME FILE '/opt/oracle/oradata/FREE/redo01.log'
                        TO '/data/oradata/FREE/redo01.log';
-- ...repeat for each redo log, then:
ALTER DATABASE OPEN RESETLOGS;
```

Also update spfile parameters that contain paths (control_files, audit_file_dest, db_recovery_file_dest) using `ALTER SYSTEM SET ... SCOPE=SPFILE` while in NOMOUNT, before restoring the control file.

> **IMPORTANT:** Files with the same name in different PDBs (e.g. system01.dbf, sysaux01.dbf in every PDB) will collide if you use a single folder with %b. Use per-PDB rules: `SET NEWNAME FOR PLUGGABLE DATABASE hrdb TO '/data/oradata/FREE/HRDB/%b';` (one line per PDB, plus root and seed).

8. Method E - Cold (Offline) File Copy

The simplest method when the target has the **same OS, same 26ai release and the same folder paths**. The database must be completely shut down during the copy.

```
-- 1. On source: note file locations (section 1.2), then
sqlplus / as sysdba
SHUTDOWN IMMEDIATE;
EXIT

# 2. Copy (Linux)
tar -czvf /u01/backup/FREE_cold.tar.gz /opt/oracle/oradata/FREE \
    $ORACLE_HOME/dbs/spfileFREE.ora $ORACLE_HOME/dbs/orapwFREE
scp /u01/backup/FREE_cold.tar.gz oracle@tgtserver:/u01/backup/

# 3. On target: stop its database, move its files aside, extract to SAME paths
sudo /etc/init.d/oracle-free-26ai stop
mv /opt/oracle/oradata/FREE /opt/oracle/oradata/FREE_orig_install
tar -xzvf /u01/backup/FREE_cold.tar.gz -C /

# 4. Start
sqlplus / as sysdba
STARTUP;
ALTER PLUGGABLE DATABASE ALL OPEN;
SHOW PDBS
```

Windows: stop the OracleServiceFREE service, copy the oradata\FREE folder plus SPFILEFREE.ORA and PWDFREE.ora from %ORACLE_HOME%\database, place them in identical paths on the target, start the service.

> **IMPORTANT:** Include **all** control files, redo logs, datafiles and tempfiles. Missing a single file means the database will not open.

9. Post-Migration Steps

9.1 Listener and service registration

```
lsnrctl status          -- should list services FREE, freepdb1, hrdb, posdb

-- If a PDB service is missing:
sqlplus / as sysdba
ALTER SYSTEM REGISTER;
SHOW PARAMETER local_listener
```

9.2 Client connection entries (tnsnames.ora)

```
HRDB =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = tgtserver)(PORT = 1521))
    (CONNECT_DATA = (SERVER = DEDICATED)(SERVICE_NAME = hrdb)))

POSDB =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = tgtserver)(PORT = 1521))
    (CONNECT_DATA = (SERVER = DEDICATED)(SERVICE_NAME = posdb)))

-- Or EZConnect, no tnsnames needed:  sqlplus hr/<pwd>@//tgtserver:1521/hrdb
```

9.3 Recompile, statistics, and fix-ups (run in each PDB)

```
ALTER SESSION SET CONTAINER = HRDB;
@?/rdbms/admin/utlrp.sql                    -- recompile invalid objects
SELECT owner, object_type, object_name FROM dba_objects WHERE status='INVALID';
EXEC DBMS_STATS.GATHER_DATABASE_STATS(degree => 1);   -- or per schema:
EXEC DBMS_STATS.GATHER_SCHEMA_STATS('HR');
```

9.4 Things that do not move automatically

- **Directory objects** - OS paths may be wrong on the new server; recreate them.
- **Database links** - host names may need updating.
- **Scheduler jobs** - check they are enabled and their OS-level scripts exist.
- **OS cron / Task Scheduler jobs**, backup scripts, and environment variables.
- **Passwords** - with Data Pump, user passwords are preserved; SYS/SYSTEM keep the target's passwords.
- **TDE wallet** - must be copied and opened if tablespaces are encrypted.

9.5 APEX / ORDS notes

- APEX lives inside each PDB. With RMAN, unplug/plug and cloning it moves automatically along with your apps.
- With Data Pump, the target PDB should have the **same APEX version** installed before importing the workspace schemas; the safest route is to also export each APEX application (App Builder > Export, or the `apex export -applicationid NNN` command in SQLcl) and the workspace, then import them on the target.
- Reconfigure ORDS on the new server to point at the new host/PDB service names.

10. Verification Checklist

Run the same queries on source and target and compare. Save the outputs to text files for evidence.

```
-- 1. PDBs open read/write, not restricted
SELECT name, open_mode, restricted FROM v$pdbs;

-- 2. Object counts per schema (run inside each PDB)
SELECT owner, object_type, COUNT(*) cnt
  FROM dba_objects
 WHERE owner IN (SELECT username FROM dba_users WHERE oracle_maintained='N')
 GROUP BY owner, object_type ORDER BY owner, object_type;

-- 3. Invalid objects (should be 0 or same as source)
SELECT owner, COUNT(*) FROM dba_objects WHERE status='INVALID' GROUP BY owner;

-- 4. Row counts for every table in a schema
SET SERVEROUTPUT ON
DECLARE n NUMBER;
BEGIN
  FOR t IN (SELECT owner, table_name FROM dba_tables WHERE owner='HR' ORDER BY 2) LOOP
    EXECUTE IMMEDIATE 'SELECT COUNT(*) FROM "'||t.owner||'"."'||t.table_name||'"' INTO n;
    DBMS_OUTPUT.PUT_LINE(RPAD(t.table_name,40)||n);
  END LOOP;
END;
/

-- 5. Users and status
SELECT username, account_status, default_tablespace FROM dba_users
 WHERE oracle_maintained='N';

-- 6. Plug-in issues
SELECT name, type, message FROM pdb_plug_in_violations WHERE status <> 'RESOLVED';
```

Check	Pass?
All PDBs listed and READ WRITE	
Object counts match source	
Row counts match on key tables	
No new invalid objects	
Application/APEX can log in and run	
Listener shows all services	
Backups configured on the new server	
Old server kept untouched until sign-off	

11. Routine Backup Strategy + Automation

Backup	Frequency	Keeps
RMAN full compressed (DATABASE PLUS ARCHIVELOG)	Daily (or weekly + daily incremental)	7-day recovery window
RMAN archivelog backup	Every few hours on busy systems	Same as above
Data Pump schema export per PDB	Daily or weekly	Last 7 dumps
Copy of backups to another disk/server/cloud	Daily	Off-machine copy
Test restore on a spare machine	Monthly	Proof the backups work

11.1 Linux - RMAN script (/home/oracle/scripts/rman_full.sh)

```
#!/bin/bash
export ORACLE_SID=FREE
export ORACLE_HOME=/opt/oracle/product/26ai/dbhomeFree    # check: echo $ORACLE_HOME
export PATH=$ORACLE_HOME/bin:$PATH
LOG=/u01/backup/logs/rman_$(date +%Y%m%d_%H%M).log
mkdir -p /u01/backup/logs

rman target / log=$LOG <<EOF
RUN {
  ALLOCATE CHANNEL c1 DEVICE TYPE DISK FORMAT '/u01/backup/rman/db_%d_%T_%U';
  BACKUP AS COMPRESSED BACKUPSET DATABASE PLUS ARCHIVELOG DELETE INPUT;
  BACKUP CURRENT CONTROLFILE FORMAT '/u01/backup/rman/ctl_%d_%T_%U';
  BACKUP SPFILE FORMAT '/u01/backup/rman/spf_%d_%T_%U';
  RELEASE CHANNEL c1;
}
CROSSCHECK BACKUP;
DELETE NOPROMPT OBSOLETE;
EOF
```

11.2 Linux - Data Pump script for all 3 PDBs (/home/oracle/scripts/dp_all.sh)

```
#!/bin/bash
export ORACLE_SID=FREE
export ORACLE_HOME=/opt/oracle/product/26ai/dbhomeFree
export PATH=$ORACLE_HOME/bin:$PATH
D=$(date +%Y%m%d)
for PDB in freepdb1 hrdb posdb; do
  expdp system/"<pwd>"@//localhost:1521/$PDB full=y directory=dp_dir \
        dumpfile=${PDB}_${D}_%U.dmp logfile=${PDB}_${D}_exp.log \
        filesize=2G flashback_time=systimestamp
done
# delete dumps older than 7 days
find /u01/backup/dp -name "*.dmp" -mtime +7 -delete
```

```
chmod 700 /home/oracle/scripts/*.sh
crontab -e
# daily RMAN at 01:00, Data Pump at 02:30
0 1 * * *  /home/oracle/scripts/rman_full.sh
30 2 * * * /home/oracle/scripts/dp_all.sh
```

11.3 Windows - RMAN (D:\scripts\rman_full.bat + rman_full.rcv)

```
REM ---- rman_full.bat ----
@echo off
set ORACLE_SID=FREE
for /f "tokens=1-3 delims=/ " %%a in ("%date%") do set D=%%c%%b%%a
rman target / cmdfile=D:\scripts\rman_full.rcv log=D:\backup\logs\rman_%D%.log

REM ---- rman_full.rcv ----
RUN {
  ALLOCATE CHANNEL c1 DEVICE TYPE DISK FORMAT 'D:\backup\rman\db_%d_%T_%U';
  BACKUP AS COMPRESSED BACKUPSET DATABASE PLUS ARCHIVELOG DELETE INPUT;
  BACKUP CURRENT CONTROLFILE FORMAT 'D:\backup\rman\ctl_%d_%T_%U';
  BACKUP SPFILE FORMAT 'D:\backup\rman\spf_%d_%T_%U';
  RELEASE CHANNEL c1;
}
CROSSCHECK BACKUP;
DELETE NOPROMPT OBSOLETE;
```

11.4 Windows - Data Pump for all PDBs (D:\scripts\dp_all.bat)

```
@echo off
set ORACLE_SID=FREE
for /f "tokens=1-3 delims=/ " %%a in ("%date%") do set D=%%c%%b%%a
for %%P in (freepdb1 hrdb posdb) do (
  expdp system/<pwd>@//localhost:1521/%%P full=y directory=dp_dir dumpfile=%%P_%D%_%%U.dmp logfile=%%P_%D%_exp.log
)
forfiles /p D:\backup\dp /m *.dmp /d -7 /c "cmd /c del @path"
```

Schedule with Task Scheduler: Create Basic Task > Daily > Start a program > the .bat file. Tick *Run whether user is logged on or not* and *Run with highest privileges*. Note: the date-parsing line depends on your Windows regional date format - test it with `echo %D%`.

> **IMPORTANT:** Avoid putting plain passwords in scripts where possible. Options: use OS authentication for RMAN (`rman target /`), restrict file permissions, or use a Secure External Password Store (wallet) for expdp.

12. Troubleshooting Common Errors

Error	Meaning	Fix
ORA-39002 / ORA-39070 / ORA-29283	Data Pump cannot write to the directory	Folder missing or no OS permission; check dba_directories path; grant READ, WRITE
ORA-39087 directory name is invalid	Directory object not created in *this* PDB	Create dp_dir inside the PDB you connect to
ORA-12514 listener does not know of service	Wrong service name or PDB closed	lsnrctl status; ALTER PLUGGABLE DATABASE x OPEN; ALTER SYSTEM REGISTER
ORA-01017 invalid username/password	Wrong password or connecting to wrong container	Check you used the PDB service, not FREE
ORA-65012 PDB already exists	Name in use on target	Drop/rename existing PDB or use another name
ORA-65122 PDB GUID conflicts	Same PDB already plugged in	Add AS CLONE to CREATE PLUGGABLE DATABASE
PDB opens RESTRICTED	Plug-in violations (patch/option mismatch)	Query pdb_plug_in_violations; run datapatch; reopen
ORA-65035 unable to create PDB from hot clone	Source not archivelog / local undo off	Enable ARCHIVELOG; check LOCAL_UNDO_ENABLED
ORA-19504 / ORA-27040 (RMAN)	Cannot create backup file	Folder missing, disk full, or permissions
RMAN-06054 unknown archived log	End of available logs during recovery	Normal - open with RESETLOGS
ORA-01113 file needs media recovery	Datafile not recovered	RECOVER DATABASE (or RECOVER DATAFILE n) before opening
ORA-19809 limit exceeded for recovery files	Recovery area full	Increase db_recovery_file_dest_size; DELETE OBSOLETE; back up and delete archivelogs
ORA-12954 request exceeds max allowed database size	Free edition 12 GB user data limit hit	Purge/shrink data, drop unused objects, or move to a licensed edition
ORA-31684 object already exists (impdp)	Object pre-exists on target	Usually ignorable on full import; or use table_exists_action

13. Quick Command Cheat Sheet

```
--- Connect ------------------------------------------------------------
sqlplus / as sysdba                              root (CDB$ROOT)
sqlplus sys/<pwd>@//localhost:1521/hrdb as sysdba  directly into a PDB
ALTER SESSION SET CONTAINER = HRDB;              switch container
SHOW CON_NAME                                    where am I?
SHOW PDBS

--- Open / close PDBs --------------------------------------------------
ALTER PLUGGABLE DATABASE hrdb OPEN;
ALTER PLUGGABLE DATABASE hrdb CLOSE IMMEDIATE;
ALTER PLUGGABLE DATABASE ALL OPEN;
ALTER PLUGGABLE DATABASE ALL SAVE STATE;

--- Data Pump ----------------------------------------------------------
expdp system/<pwd>@//localhost:1521/hrdb full=y directory=dp_dir dumpfile=x_%U.dmp logfile=x.log
impdp system/<pwd>@//localhost:1521/hrdb full=y directory=dp_dir dumpfile=x_%U.dmp logfile=i.log
expdp ... schemas=HR       impdp ... remap_schema=HR:HR2  remap_tablespace=USERS:HR_DATA

--- Unplug / plug ------------------------------------------------------
ALTER PLUGGABLE DATABASE hrdb CLOSE IMMEDIATE;
ALTER PLUGGABLE DATABASE hrdb UNPLUG INTO '/u01/backup/pdb/hrdb.pdb';
CREATE PLUGGABLE DATABASE hrdb USING '/u01/backup/pdb/hrdb.pdb' CREATE_FILE_DEST='/path';

--- Remote clone -------------------------------------------------------
CREATE PLUGGABLE DATABASE hrdb FROM hrdb@clone_link FILE_NAME_CONVERT=('src/','tgt/');

--- RMAN ---------------------------------------------------------------
rman target /
BACKUP AS COMPRESSED BACKUPSET DATABASE PLUS ARCHIVELOG;
BACKUP PLUGGABLE DATABASE hrdb;
LIST BACKUP SUMMARY;      RESTORE DATABASE VALIDATE;
CROSSCHECK BACKUP;        DELETE NOPROMPT OBSOLETE;
RESTORE PLUGGABLE DATABASE hrdb;  RECOVER PLUGGABLE DATABASE hrdb;   (PDB closed)

--- Service (Linux) ----------------------------------------------------
sudo /etc/init.d/oracle-free-26ai start | stop | status
lsnrctl status | start | stop
```

Golden rules: (1) same release on both sides for physical moves; (2) verify every log file says *successfully completed*; (3) test the restore before switching users over; (4) keep the old server untouched until the new one is signed off.